AUDIO-VIDEO IMPLEMENTATION

TRAINING

```
Feature dims -> audio: 28, video: 2304
Epoch 01 | train_loss 1.3070 | val_MAE 1.4061 | val_Acc2 0.6026 | val_F1 0.4577 | val_Corr 0.0
660
Epoch 02 | train_loss 1.2685 | val_MAE 1.4106 | val_Acc2 0.5808 | val_F1 0.5662 | val_Corr 0.0
558
Epoch 03 | train_loss 1.2335 | val_MAE 1.4263 | val_Acc2 0.5808 | val_F1 0.5606 | val_Corr 0.0
426
Epoch 04 | train_loss 1.1923 | val_MAE 1.4621 | val_Acc2 0.5852 | val_F1 0.4562 | val_Corr 0.0
411
Epoch 05 | train_loss 1.1707 | val_MAE 1.4481 | val_Acc2 0.5590 | val_F1 0.5156 | val_Corr 0.0
074
Epoch 06 | train_loss 1.1374 | val_MAE 1.4108 | val_Acc2 0.5895 | val_F1 0.5506 | val_Corr 0.0
979
Epoch 07 | train_loss 1.0950 | val_MAE 1.4827 | val_Acc2 0.5371 | val_F1 0.5293 | val_Corr -0.
0187
Epoch 08 | train_loss 1.0786 | val_MAE 1.4038 | val_Acc2 0.5939 | val_F1 0.5446 | val_Corr 0.1
817
Epoch 09 | train_loss 1.0480 | val_MAE 1.4189 | val_Acc2 0.5808 | val_F1 0.5808 | val_Corr 0.1
087
Epoch 10 | train_loss 1.0348 | val_MAE 1.4155 | val_Acc2 0.5939 | val_F1 0.5919 | val_Corr 0.1
276
Epoch 11 | train_loss 1.0222 | val_MAE 1.4721 | val_Acc2 0.5371 | val_F1 0.5416 | val_Corr 0.0
601
Epoch 12 | train_loss 1.0174 | val_MAE 1.4339 | val_Acc2 0.5852 | val_F1 0.5788 | val_Corr 0.0
894
Epoch 13 | train_loss 1.0096 | val_MAE 1.4116 | val_Acc2 0.5633 | val_F1 0.5463 | val_Corr 0.1
354
Epoch 14 | train_loss 0.9672 | val_MAE 1.4000 | val_Acc2 0.5895 | val_F1 0.5895 | val_Corr 0.1
472
Epoch 15 | train_loss 0.9656 | val_MAE 1.4571 | val_Acc2 0.5895 | val_F1 0.5799 | val_Corr 0.0
902
Epoch 16 | train_loss 0.9376 | val_MAE 1.5080 | val_Acc2 0.4978 | val_F1 0.4925 | val_Corr 0.1
036
```

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Epoch 17 | train_loss 0.9410 | val_MAE 1.4294 | val_Acc2 0.5808 | val_F1 0.5564 | val_Corr 0.1
314
Epoch 18 | train_loss 0.9331 | val_MAE 1.4570 | val_Acc2 0.5721 | val_F1 0.5745 | val_Corr 0.0
954
Epoch 19 | train_loss 0.9194 | val_MAE 1.4413 | val_Acc2 0.5721 | val_F1 0.5620 | val_Corr 0.1
261
Epoch 20 | train_loss 0.9006 | val_MAE 1.4599 | val_Acc2 0.5721 | val_F1 0.5740 | val_Corr 0.0
970
Epoch 21 | train_loss 0.8900 | val_MAE 1.4485 | val_Acc2 0.5895 | val_F1 0.5850 | val_Corr 0.1
084
Epoch 22 | train_loss 0.8896 | val_MAE 1.4570 | val_Acc2 0.5808 | val_F1 0.5815 | val_Corr 0.1
010
Epoch 23 | train_loss 0.8816 | val_MAE 1.4619 | val_Acc2 0.5808 | val_F1 0.5808 | val_Corr 0.1
032
Epoch 24 | train_loss 0.8874 | val_MAE 1.4647 | val_Acc2 0.5677 | val_F1 0.5552 | val_Corr 0.0
900
Epoch 25 | train_loss 0.8725 | val_MAE 1.4656 | val_Acc2 0.5633 | val_F1 0.5585 | val_Corr 0.0
905
Epoch 26 | train_loss 0.8779 | val_MAE 1.4685 | val_Acc2 0.5764 | val_F1 0.5733 | val_Corr 0.0
854
Epoch 27 | train_loss 0.8592 | val_MAE 1.4731 | val_Acc2 0.5721 | val_F1 0.5704 | val_Corr 0.0
835
Epoch 28 | train_loss 0.8633 | val_MAE 1.4736 | val_Acc2 0.5677 | val_F1 0.5673 | val_Corr 0.0
843
Epoch 29 | train_loss 0.8581 | val_MAE 1.4808 | val_Acc2 0.5633 | val_F1 0.5653 | val_Corr 0.0
805
Epoch 30 | train_loss 0.8490 | val_MAE 1.4728 | val_Acc2 0.5808 | val_F1 0.5772 | val_Corr 0.0
871
```

```
Epoch 29 | train_loss 0.8581 | val_MAE 1.4808 | val_Acc2 0.5633 | val_F1 0.5653 | val_Corr 0.0
805
Epoch 30 | train_loss 0.8490 | val_MAE 1.4728 | val_Acc2 0.5808 | val_F1 0.5772 | val_Corr 0.0
871
Epoch 31 | train_loss 0.8549 | val_MAE 1.4845 | val_Acc2 0.5677 | val_F1 0.5681 | val_Corr 0.0
727
Epoch 32 | train_loss 0.8522 | val_MAE 1.5005 | val_Acc2 0.5459 | val_F1 0.5490 | val_Corr 0.0
641
Epoch 33 | train_loss 0.8485 | val_MAE 1.4921 | val_Acc2 0.5590 | val_F1 0.5613 | val_Corr 0.0
704
Epoch 34 | train_loss 0.8537 | val_MAE 1.4860 | val_Acc2 0.5721 | val_F1 0.5694 | val_Corr 0.0
757
Epoch 35 | train_loss 0.8503 | val_MAE 1.4902 | val_Acc2 0.5677 | val_F1 0.5673 | val_Corr 0.0
736
Epoch 36 | train_loss 0.8455 | val_MAE 1.4920 | val_Acc2 0.5590 | val_F1 0.5593 | val_Corr 0.0
705
Epoch 37 | train_loss 0.8435 | val_MAE 1.4928 | val_Acc2 0.5677 | val_F1 0.5673 | val_Corr 0.0
702
Epoch 38 | train_loss 0.8371 | val_MAE 1.4909 | val_Acc2 0.5677 | val_F1 0.5673 | val_Corr 0.0
715
Epoch 39 | train_loss 0.8377 | val_MAE 1.4948 | val_Acc2 0.5502 | val_F1 0.5526 | val_Corr 0.0
685
Epoch 40 | train_loss 0.8374 | val_MAE 1.4908 | val_Acc2 0.5546 | val_F1 0.5553 | val_Corr 0.0
709

=== TEST RESULTS ===
MAE: 1.6510
Corr: -0.0840
Acc2: 0.4854
F1: 0.4868
```

```
Dims -> audio: 28, video: 2304

============================================================
Training FULL model (Adaptive Gate)...
============================================================
[adaptive] Epoch 10 | val_MAE 1.3236 | val_Acc2 0.6463
[adaptive] Epoch 20 | val_MAE 1.3438 | val_Acc2 0.6419
[adaptive] Epoch 30 | val_MAE 1.3778 | val_Acc2 0.5939
[adaptive] Epoch 40 | val_MAE 1.3702 | val_Acc2 0.6157

============================================================
Training ABLATION baseline (Fixed/Average Fusion, no adaptive gate)...
============================================================
[fixed] Epoch 10 | val_MAE 1.3575 | val_Acc2 0.6157
[fixed] Epoch 20 | val_MAE 1.4410 | val_Acc2 0.5939
[fixed] Epoch 30 | val_MAE 1.4907 | val_Acc2 0.5895
[fixed] Epoch 40 | val_MAE 1.4919 | val_Acc2 0.5983

============================================================
ABLATION RESULTS (same data, same encoders/attention, only fusion step differs)
============================================================
Metric     Adaptive Gate   Fixed (Avg) Fusion   Improvement
MAE        1.6996          1.7037               +0.0041 (lower is better)
Corr       -0.0888         -0.0500              -0.0388 (higher is better)
Acc2       0.4781          0.4548               +0.0233 (higher is better)
F1         0.4731          0.4324               +0.0407 (higher is better)

Adaptive gate average weights on TEST set:
  Audio: 0.6394
  Video: 0.3606

(Gate weights show what the gate used, not which modality is more informative.)
```

EVALUATION AND TESTING –

```
Modalities: ('audio', 'video') | device: cuda | data: /kaggle/input/datasets/aakarsonawane/adaptive
-fusion/mosi_features/mosi_data.pkl | checkpoint: /kaggle/input/datasets/aakarsonawane/adaptive-fus
ion/best_model.pt

================================================================================
A. EDA
================================================================================
train  n=1283  mean=+0.235 std=1.510 | pos 679 zero 52 neg 552 | majority Acc-2 baseline 0.5698
valid  n=229   mean=+0.310 std=1.649 | pos 124 zero 13 neg 92 | majority Acc-2 baseline 0.5983
test   n=686   mean=-0.317 std=1.588 | pos 277 zero 30 neg 379 | majority Acc-2 baseline 0.5525
audio  dim=28    mean=-12.506 std=72.357 range=[-798.82,279.35] | all-zero frames 1.8% | samples >5
0% zero frames 0.3%
video  dim=2304  mean=+0.271 std=0.250 range=[0.00,1.00] | all-zero frames 28.1% | samples >50% zer
o frames 25.9%
  (video all-zero frames = padding + frames where no face was detected)

================================================================================
B. STATISTICAL EVALUATION (test split)
================================================================================
Metric        Point               95% CI
MAE          1.6510   [1.5697, 1.7382]
Corr        -0.0840   [-0.1563, -0.0097]
Acc2         0.4854   [0.4475, 0.5248]
Acc2_no0     0.4848   [0.4458, 0.5237]
F1           0.4868   [0.4495, 0.5252]
Acc7         0.1603   [0.1327, 0.1866]

Majority-class Acc-2 baseline: 0.5525 | model - baseline = -0.0671 (95% CI [-0.1050, -0.0277])  ->
NOT clearly above baseline
```

```
Classification report (0 = negative, 1 = non-negative):
              precision    recall  f1-score   support

    negative     0.5380    0.4855    0.5104       379
non-negative     0.4331    0.4853    0.4578       307

    accuracy                         0.4854       686
   macro avg     0.4856    0.4854    0.4841       686
weighted avg     0.4911    0.4854    0.4868       686

Confusion matrix [[TN FP],[FN TP]]:
 [[184 195]
 [158 149]]

Top-10 largest errors (sample index, true, pred):
  #158  true -3.00  pred +1.95  |err| 4.95
  #6    true -3.00  pred +1.78  |err| 4.78
  #257  true -2.60  pred +2.12  |err| 4.72
  #144  true -2.80  pred +1.85  |err| 4.65
  #64   true +3.00  pred -1.53  |err| 4.53
  #127  true -2.80  pred +1.62  |err| 4.42
  #203  true -2.60  pred +1.82  |err| 4.42
  #154  true -3.00  pred +1.41  |err| 4.41
  #146  true -2.60  pred +1.75  |err| 4.35
  #139  true -2.80  pred +1.54  |err| 4.34

MAE by true-sentiment bin:
  [-3, -2)  n=96   MAE=2.812
  [-2, -1)  n=167  MAE=1.853
  [-1, +0)  n=116  MAE=0.981
  [+0, +1)  n=124  MAE=1.000
  [+1, +2)  n=104  MAE=1.367
  [+2, +3)  n=79   MAE=2.192
```

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================================================================================
C. ROBUSTNESS (accuracy drop, paired bootstrap 95% CI)
================================================================================
(noise conditions averaged over 5 random draws; noise is added only to non-padding frames)
  done: Audio noise 0.15x
  done: Audio noise 0.5x
  done: Audio noise 1.0x
  done: Audio noise 2.0x
  done: Audio truncated (keep 50%)
  done: Audio truncated (keep 25%)
  done: Video element dropout 20%
  done: Video element dropout 50%
  done: Video element dropout 80%
  done: Video frame dropout 30% (no face)
  done: Video frame dropout 60% (no face)
  done: Audio+video noise 1.0x
  done: Audio zeroed
  done: Video zeroed

Condition                              Acc%    Drop              95% CI  CI excl. 0  Holm p<.05
Audio noise 0.15x                     48.72   -0.17   [ -1.08, +0.67]          no          no
Audio noise 0.5x                      47.93   +0.61   [ -0.88, +2.13]          no          no
Audio noise 1.0x                      47.52   +1.02   [ -1.08, +3.15]          no          no
Audio noise 2.0x                      47.64   +0.90   [ -1.46, +3.53]          no          no
Audio truncated (keep 50%)            49.42   -0.87   [ -3.64, +2.19]          no          no
Audio truncated (keep 25%)            49.42   -0.87   [ -4.66, +3.06]          no          no
Video element dropout 20%             48.51   +0.03   [ -2.71, +2.95]          no          no
Video element dropout 50%             50.73   -2.19   [ -6.62, +2.39]          no          no
Video element dropout 80%             54.02   -5.48   [-10.00, -1.22]         yes          no
Video frame dropout 30% (no face)     49.77   -1.22   [ -3.00, +0.61]          no          no
Video frame dropout 60% (no face)     50.61   -2.07   [ -4.69, +0.18]          no          no
Audio+video noise 1.0x                47.87   +0.67   [ -1.66, +2.77]          no          no
Audio zeroed                          52.92   -4.37   [ -8.60, -0.15]         yes          no
Video zeroed                          44.75   +3.79   [ -1.02, +8.89]          no          no
```

```
Mean gate weights (audio, video):
  Audio noise 0.15x                    0.534  0.466
  Audio noise 0.5x                     0.538  0.462
  Audio noise 1.0x                     0.553  0.447
  Audio noise 2.0x                     0.594  0.406
  Audio truncated (keep 50%)           0.643  0.357
  Audio truncated (keep 25%)           0.711  0.289
  Video element dropout 20%            0.592  0.408
  Video element dropout 50%            0.659  0.341
  Video element dropout 80%            0.620  0.380
  Video frame dropout 30% (no face)    0.560  0.440
  Video frame dropout 60% (no face)    0.548  0.452
  Audio+video noise 1.0x               0.569  0.431
  Audio zeroed                         0.740  0.260
  Video zeroed                         0.313  0.687

================================================================================
D. UNIT TESTS
================================================================================
  [PASS] output shapes (pred, gate)
  [PASS] outputs finite
  [PASS] gate weights sum to 1
  [PASS] eval-mode determinism
  [FAIL] batch output == single-sample output ValueError: zero-dimensional arrays cannot be concate
nated
  [PASS] all-zero modality / all-zero input stays finite
  [PASS] very short (T=1) and long (T=100) sequences
  [PASS] NaN/inf sanitized by predict_api
  [PASS] wrong feature dimension is rejected
  [PASS] checkpoint save/load round trip
  [PASS] predict_api == batch inference
  [PASS] predict_api rejects malformed input

  11/12 passed
```

```
================================================================================
E. EFFICIENCY
================================================================================
Parameters: 3,010,371 | checkpoint size: 12.06 MB

Forward-pass latency, CPU (2 threads):
  batch 1   p50     8.22 ms  p95     9.65  p99     9.89     119.1 samples/s
  batch 8   p50    33.54 ms  p95    36.54  p99    37.52     237.2 samples/s
  batch 32  p50    86.91 ms  p95    95.72  p99   123.04     360.0 samples/s
  batch 64  p50   175.81 ms  p95   189.14  p99   191.11     360.9 samples/s

Forward-pass latency, GPU:
  batch 1   p50     2.66 ms  p95     2.72  p99     3.88     389.8 samples/s
  batch 8   p50     2.49 ms  p95     2.85  p99     2.95    3152.7 samples/s
  batch 32  p50     3.90 ms  p95     3.99  p99     4.02    8389.9 samples/s
  batch 64  p50     7.45 ms  p95     7.65  p99     7.67    8633.0 samples/s

================================================================================
F. END-TO-END PIPELINE LATENCY (raw clip -> features -> model)
================================================================================
10 random raw clips, CPU (2 threads), one clip at a time:
  clip duration (s)                        median      3.35   p95      9.29
  audio features (ms)                      median     44.93   p95  10449.92
  face detection + video features (ms)     median   1427.59   p95   4282.22
  model forward (ms)                       median     11.04   p95     15.93
  TOTAL (ms)                               median   1494.85   p95  14542.96
  real-time factor (total / clip duration) median      0.52   p95      2.43

================================================================================
H. TABLES TO PASTE INTO THE REPORT
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================================================================================
H. TABLES TO PASTE INTO THE REPORT
================================================================================

**Table 1. Test-set performance (95% bootstrap CI)**

| Metric | Value | 95% CI |
|---|---|---|
| MAE | 1.6510 | [1.5697, 1.7382] |
| Corr | -0.0840 | [-0.1563, -0.0097] |
| Acc2 | 0.4854 | [0.4475, 0.5248] |
| Acc2_no0 | 0.4848 | [0.4458, 0.5237] |
| F1 | 0.4868 | [0.4495, 0.5252] |
| Acc7 | 0.1603 | [0.1327, 0.1866] |
| Majority Acc-2 baseline | 0.5525 | n/a |

**Table 2. Robustness (accuracy drop in points, paired bootstrap CI)**

| Condition | Acc (%) | Drop | 95% CI | CI excludes 0 | Holm p<0.05 |
|---|---|---|---|---|---|
| Audio noise 0.15x | 48.72 | -0.17 | [-1.08, +0.67] | no | no |
| Audio noise 0.5x | 47.93 | +0.61 | [-0.88, +2.13] | no | no |
| Audio noise 1.0x | 47.52 | +1.02 | [-1.08, +3.15] | no | no |
| Audio noise 2.0x | 47.64 | +0.90 | [-1.46, +3.53] | no | no |
| Audio truncated (keep 50%) | 49.42 | -0.87 | [-3.64, +2.19] | no | no |
| Audio truncated (keep 25%) | 49.42 | -0.87 | [-4.66, +3.06] | no | no |
| Video element dropout 20% | 48.51 | +0.03 | [-2.71, +2.95] | no | no |
| Video element dropout 50% | 50.73 | -2.19 | [-6.62, +2.39] | no | no |
| Video element dropout 80% | 54.02 | -5.48 | [-10.00, -1.22] | yes | no |
| Video frame dropout 30% (no face) | 49.77 | -1.22 | [-3.00, +0.61] | no | no |
| Video frame dropout 60% (no face) | 50.61 | -2.07 | [-4.69, +0.18] | no | no |
| Audio+video noise 1.0x | 47.87 | +0.67 | [-1.66, +2.77] | no | no |
| Audio zeroed | 52.92 | -4.37 | [-8.60, -0.15] | yes | no |
| Video zeroed | 44.75 | +3.79 | [-1.02, +8.89] | no | no |

**Table 3. Efficiency** (params 3,010,371, checkpoint 12.06 MB)

| Device | Batch | p50 (ms) | p95 (ms) | p99 (ms) | Samples/s |
|---|---|---|---|---|---|
| CPU | 1 | 8.22 | 9.65 | 9.89 | 119.1 |
| CPU | 8 | 33.54 | 36.54 | 37.52 | 237.2 |
| CPU | 32 | 86.91 | 95.72 | 123.04 | 360.0 |
| CPU | 64 | 175.81 | 189.14 | 191.11 | 360.9 |
| GPU | 1 | 2.66 | 2.72 | 3.88 | 389.8 |
| GPU | 8 | 2.49 | 2.85 | 2.95 | 3152.7 |
| GPU | 32 | 3.90 | 3.99 | 4.02 | 8389.9 |
| GPU | 64 | 7.45 | 7.65 | 7.67 | 8633.0 |

**Table 4. Unit tests: 11/12 passed**

Saved results -> /kaggle/working/eval_outputs/eval_results.json
Figures in /kaggle/working/eval_outputs: label_distribution.png, stats_figures.png, robustness.png
```